

Airbnb Dynamic Pricing Recommendation Engine

Tableau Dashboard – Screenshot and Explanation

This document explains the Tableau dashboard shown in the uploaded screenshot. The dashboard is designed to analyze Airbnb pricing, ratings, room types and property characteristics as part of the Dynamic Pricing Recommendation Engine project.

1. Dashboard Overview

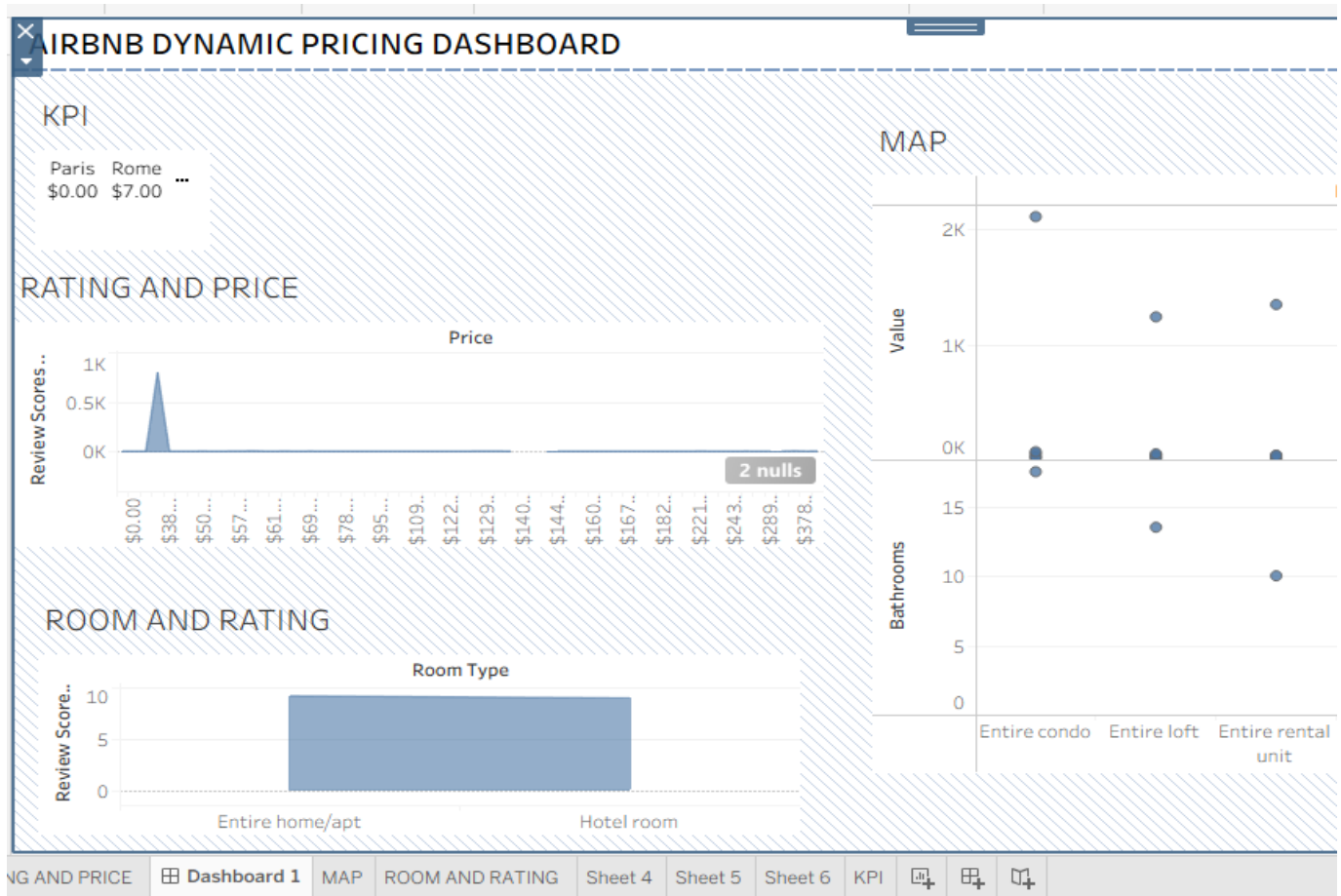


Figure 1: Airbnb Dynamic Pricing Dashboard created in Tableau.

2. Dashboard Title

The title 'AIRBNB DYNAMIC PRICING DASHBOARD' identifies the purpose of the dashboard. It brings together different visualizations that help understand Airbnb listing prices and related factors.

3. KPI Section

The KPI section provides a quick summary of important values. In the screenshot, values such as Paris and Rome are visible with dollar amounts. KPIs are useful because they allow the user to view important pricing information quickly without examining every individual listing.

4. Rating and Price

The 'RATING AND PRICE' visualization compares review-score information with Airbnb price. The horizontal axis contains price values and the vertical axis represents review-score values. This chart can be used to investigate whether higher-rated listings are associated with different prices.

5. Room and Rating

The 'ROOM AND RATING' chart compares room types with review scores. The room types visible include 'Entire home/apt' and 'Hotel room'. This visualization helps compare listing quality across accommodation categories.

6. Map Section

The 'MAP' section provides a geographical view of the Airbnb data. Each plotted point represents a location or property-related observation. A map is useful for identifying geographical patterns in Airbnb listings and pricing.

7. Property Characteristics in the Map/Chart Area

The right side of the dashboard also displays property characteristics such as bathrooms and room/property categories. These visual elements help users examine whether property size or type is associated with different values in the dataset.

8. Dashboard Components

Component	Purpose
KPI	Shows key pricing or summary indicators.
Rating and Price	Examines the relationship between price and review score.
Room and Rating	Compares review scores across room types.
Map	Shows geographical distribution of Airbnb observations.
Property/Bathroom View	Helps compare property characteristics.

9. '2 nulls' Indicator

The '2 nulls' indicator visible on the Rating and Price chart means that two records contain missing values for the fields used by that visualization. These missing values should be checked during data cleaning so that they do not affect the analysis.

10. Project Connection

The Tableau dashboard is the visualization stage of the project. The cleaned Airbnb data and Python pricing model can provide predicted prices, which can then be displayed in Tableau. Filters can be added for city, property type, room type and other variables so that users can explore pricing patterns.

11. Recommended Final Dashboard Flow

Stage	Output
Excel	Initial data preparation and summary
Python	Data cleaning, analysis and regression prediction
Pricing Engine	Recommended/dynamic price
Tableau	Interactive dashboard and visual analysis
Final PDF	Project findings and recommendations

Conclusion

The Tableau dashboard provides an interactive visual summary of Airbnb pricing and listing characteristics. It combines KPIs, rating-price analysis, room-type comparisons and geographical information. For the final Dynamic Pricing Recommendation Engine, the dashboard can be connected to the Python-generated predicted-price dataset and enhanced with filters and a price suggestion control.